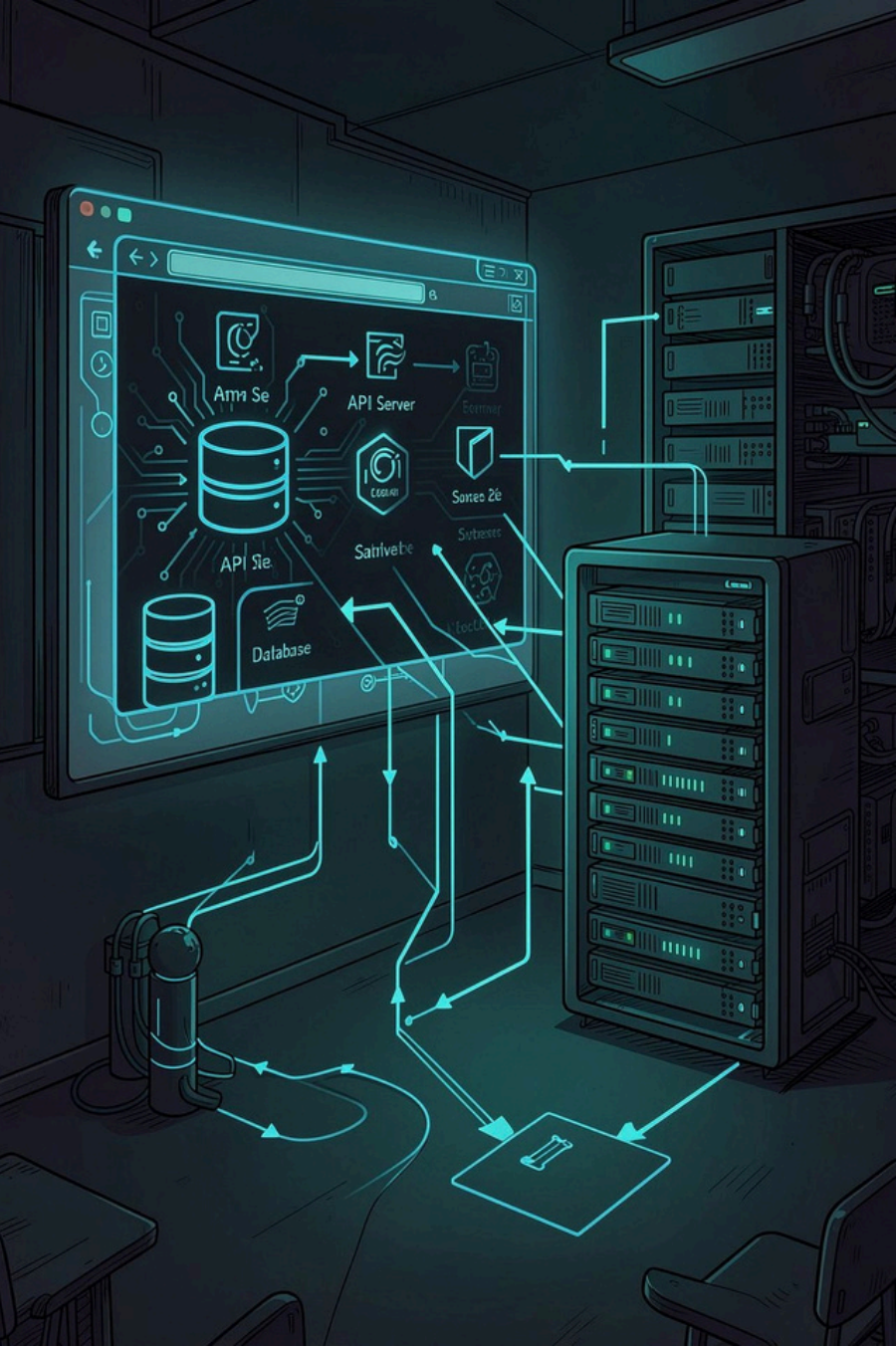
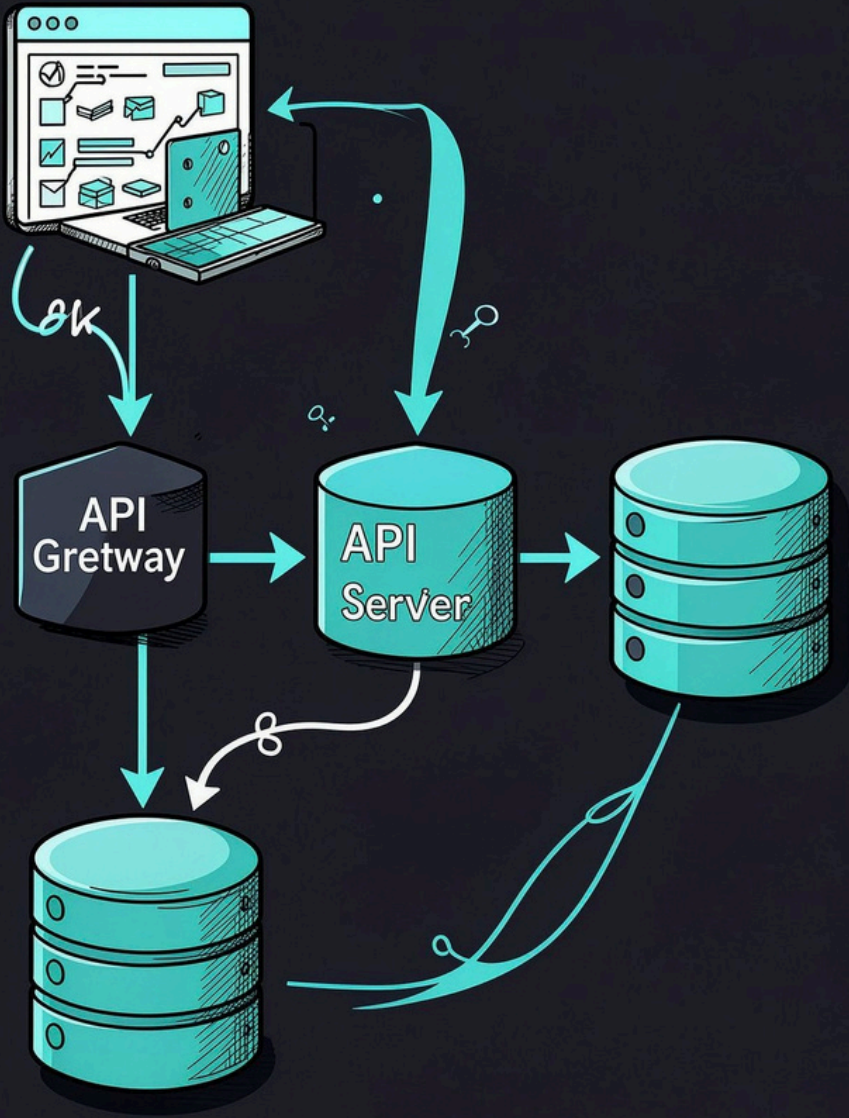


API Integration in Web Technology

Connecting Web Applications with External Services

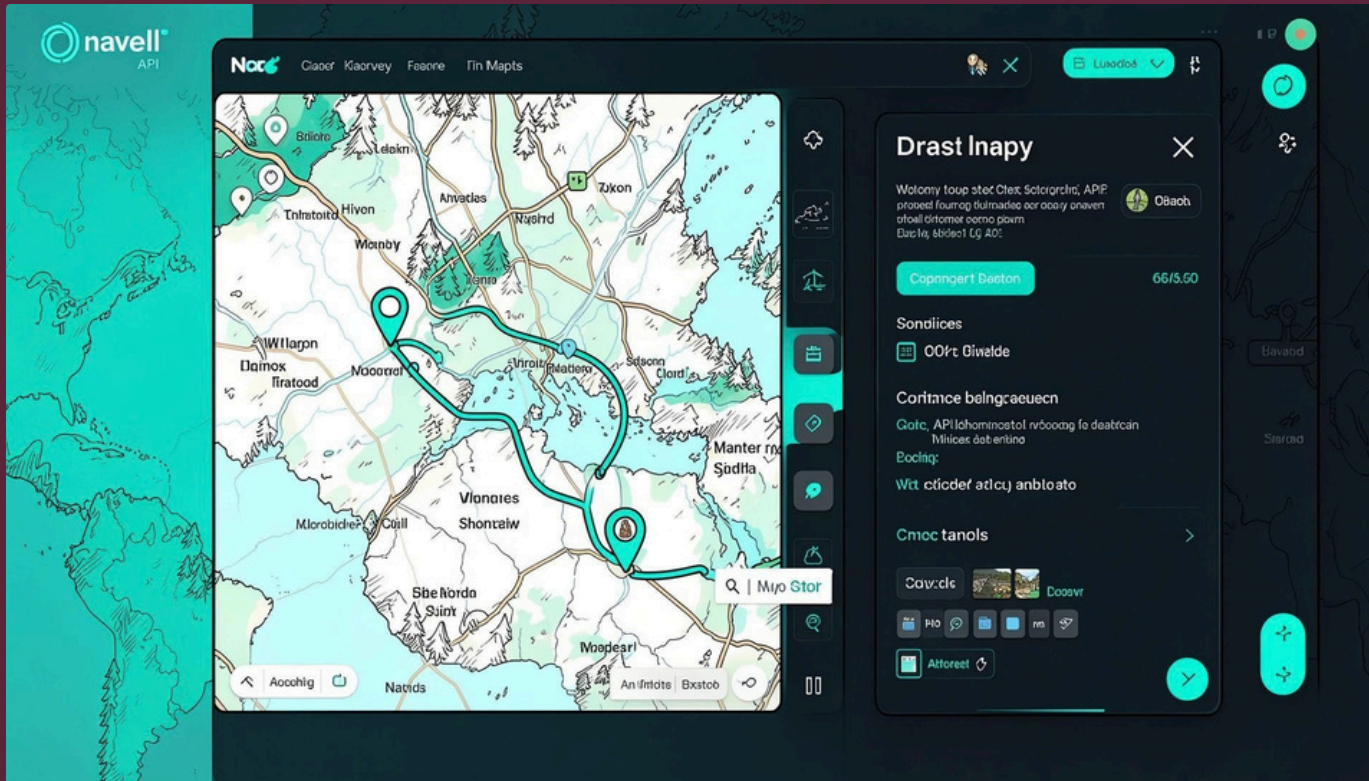




What is an API?

- **API**=Application Programming Interface
- Bridge for two software systems to **request** and **exchange** data
- Commonly uses **HTTP/HTTPS** and **JSON**
- Real-world example: Weather app → **Weather API** → weather data

What is API Integration?



API integration = connecting an app to external services via APIs

- Send requests → Receive data
- Access external services (maps, payments, auth)
- Automate cross-system communication

📄 Example: Travel Website → Maps API → Location & map data

Why API Integration Matters



Faster development

Reuse existing services → ship features quicker



System connectivity

Connect heterogeneous platforms



Real-time data

Live updates from external sources

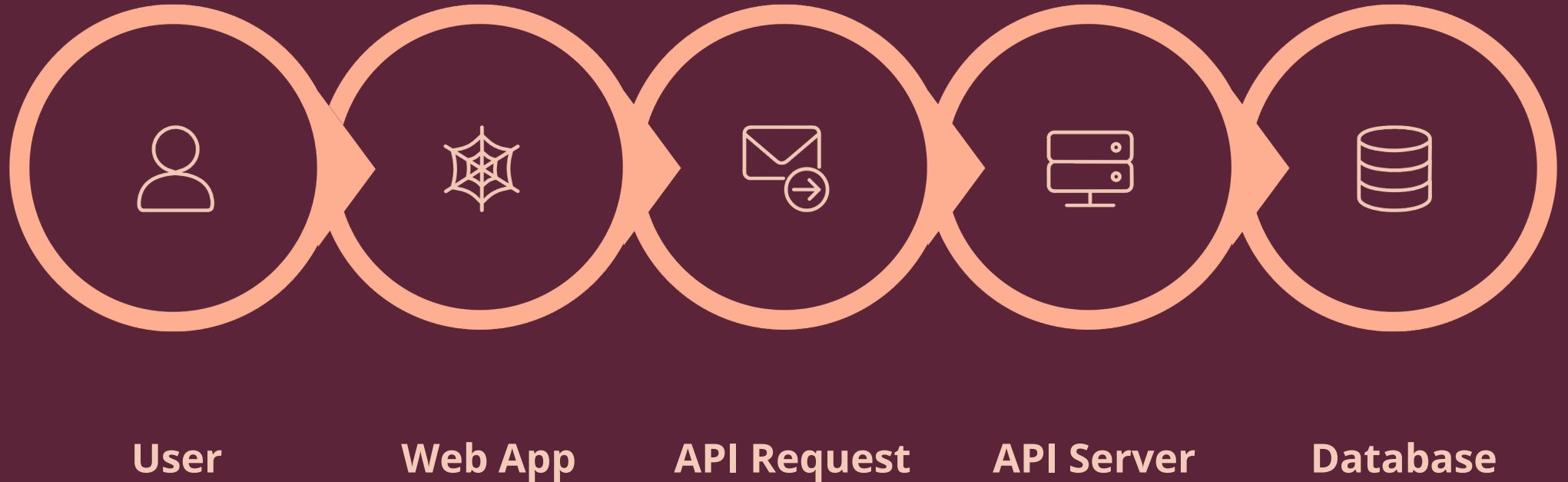


Cloud & payments

Integrate auth, billing, storage

Developers use services instead of building everything from scratch.

How API Integration Works



A user action → app sends API request → server processes → returns data → UI updates

HTTP Methods (Memory Guide)

HTTP Method	Purpose	Example
GET	Retrieve data	Get user details
POST	Create data	Create account
PUT	Replace/update	Update profile
PATCH	Partial update	Change email
DELETE	Remove data	Delete account

Short mnemonic: GET = Read · POST = Create · PUT/PATCH = Update · DELETE = Remove

API Request & Response

```
GET /users/101 HTTP/1.1  
Host: api.example.com  
Accept: application/json
```

```
{  
  "id": 101,  
  "name": "Angelina",  
  "course": "CSE AIML"  
}
```

- Client sends **request** (endpoint + method)
- Server returns **response** (status + body)
- **JSON** is the common data format

REST API — Key Principles

1 Resources via URLs

Endpoints identify resources (e.g., /students/101)

2 Stateless

Each request contains all necessary info

3 HTTP methods

Use GET, POST, PUT, PATCH, DELETE

4 JSON payloads

Lightweight and easy to parse

Examples: GET /students · POST /students · PUT /students/101 · DELETE /students/101

API Integration in JavaScript (Fetch)

```
fetch("https://api.example.com/users")  
  .then(res => res.json())  
  .then(data => console.log(data))  
  .catch(err => console.log(err));
```

- `fetch()` sends request
- `response.json()` parses JSON → usable JS object
- Use `try/catch` or `.catch` for errors

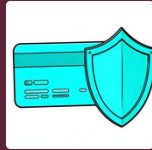


Real-World API Examples



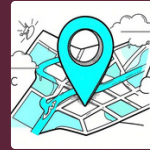
Weather APIs

Temperature, humidity, forecasts



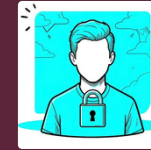
Payment APIs

Payments, transaction verification



Maps APIs

Locations, directions, distances



Authentication APIs

OAuth, social login (Google, GitHub)